CARDIFF UNIVERSITY
PRIFYSGOL CAERDYDD

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Academic Year: 2022/23

Examination Period: Spring

Module Code: MAT700

Examination Paper Title: Mathematical Methods for Data Mining

Duration: 2 hours

Please read the following information carefully:

Structure of Examination Paper:

- There are 4 pages including this page.
- There are **4** questions in total.
- There are no appendices.
- The maximum mark for the examination paper is 75 and the mark obtainable for a question or part of a question is shown in brackets alongside the question.

Instructions for completing the examination:

- Complete the front cover of any answer books used.
- This examination paper must be submitted to an Invigilator at the end of the examination.
- Answer **THREE** questions.
- Each question should be answered on a separate page.

You will be provided with / or allowed:

- **ONE** answer book.
- Squared graph paper.
- The **use of calculators** is **not permitted** in this examination.
- The use of a translation dictionary between English or Welsh and another language, provided that it bears an appropriate departmental stamp, is permitted in this examination.
- The use of the student's own notes, up to **1 sheet (2 sides) of A4 paper**, is permitted in this examination.

1. (a) What are supervised learning, unsupervised learning and semi-supervised learning? Give examples. [6]

   (b) i. Explain Bonferroni's principle for Data Mining. [2]

   ii. In a class of 150 students, suppose that the birthdays are independent and uniformly distributed over the 365 days of the year. We are interested in the event that some students have their birthday on the same day. Calculate the expectation of the number of triples of students having the same birthday. [4]

   (c) i. Define the TF×IDF score for words in a collection of documents. [3]

   ii. Show that in a corpus of 500 documents, any word in any document has a TF×IDF score less than 9. [3]

   iii. For the following document-term matrix for a corpus of documents:

	T1	T2	T3	T4	T5	T6
Doc1	2	0	4	3	0	1
Doc2	0	2	4	0	2	3
Doc3	4	0	1	3	0	1
Doc4	0	1	0	2	0	0
Doc5	0	0	2	0	0	4

   calculate the corresponding vector representation of the documents (TF×IDF matrix). [4]

   (d) Prove that the *edit distance* is indeed a distance measure. [3]

2. (a) i. Describe how to calculate precision, recall and F-measure of a binary classifier. [6]

   ii. Prove that F-measure of any binary classifier is $\leq \frac{precision+recall}{2}$. [4]

   iii. During testing of a Spam/NotSpam classifier on a test set of 500 e-mail messages with 100 spam messages and 400 non-spam messages you found that 90 spam messages were correctly classified as spam, and 300 non-spam messages were correctly classified as not spam. Calculate the precision, recall and F-measure of this classifier. [6]

   (b) i. Describe how to calculate the Google PageRank function. [4]

   ii. Prove that any two web-pages with only out-links have the same PageRank value. [5]

3. (a) What is the kNN classification algorithm? Explain how to weight the contributions of the neighbours. [4]

(b) Consider the following data:

Instance	X_1	X_2	Class
1	0.25	0.25	+
2	0.25	0.75	+
3	0.50	0.25	-
4	0.50	0.75	-
5	0.75	0.50	-
6	0.75	1.00	+
7	0.25	0.80	?

How does the weighted version of kNN algorithm classify instance 7, with k = 4 and L_∞ distance? [5]

(c) i. Describe the Naïve Bayesian classification. [5]

ii. Given the following data,

Instance	A	B	C	Class
1	0	0	0	+
2	0	0	1	-
3	0	1	1	-
4	0	1	1	-
5	0	0	1	+
6	1	0	1	+
7	1	0	0	-
8	1	0	1	-
9	1	1	1	+
10	1	0	1	+

predict the class label for instance (A = 1; B = 1; C = 0) using Naïve Bayes classification. [5]

(d) Formulate the k-means algorithm. What is the objective function of the k-means algorithm? [6]

4. (a) i. Explain the following terms: linearly separable data, margin and support vectors. [3]

ii. For a given data set of 3-D input data, we apply the SVM learning algorithm and therefore achieve an optimal decision plane:

$$x_1 - 2x_2 + 12x_3 - 33 = 0.$$

What is the margin of this SVM? [3]

iii. Explain how to reduce a single multiclass classification problem into multiple binary classification problems. [4]

(b) i. What is the unnormalized graph Laplacian operator L for a weighted graph? [2]

ii. Define a harmonic function for a weighted graph with labelled data. [3]

(c) i. What is a meta-algorithm? Describe the halving algorithm and the weighted majority algorithm. [5]

ii. Suppose we have access to a set of n experts and assume that one of the experts is consistent. Prove that the halving algorithm makes at most $\log_2(n)$ mistakes. [5]